

Pradeep Pujari

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◆ Professional Summary

Innovative and results-driven AI/ML Engineer with deep expertise in Deep Neural Networks, Machine Learning, Natural Language Processing (NLP), Generative AI (LLMs), and Search Science. Specialized in Large Language Models, Agentic AI, and alignment techniques including RLHF, DPO, and GRPO. Proven ability to design and deploy scalable, customer-centric AI solutions by translating cutting-edge research into production-ready systems with measurable impact. Strong collaborator with a focus on system performance, reproducibility, and cross-functional product delivery.

◆ Core Competencies

- **AI & ML Expertise:** Machine Learning, Deep Neural Networks, Reinforcement Learning, Large Language Models (LLMs), Natural Language Understanding (NLU), Computer Vision, Graph Neural Networks (GNNs), Transformers, Knowledge Graphs, Information Retrieval
- **Model Optimization & Deployment:** Scalable model training, hyperparameter tuning, inference optimization, data and model parallelism
- **MLOps & Automation:** Pipeline automation, experiment reproducibility, monitoring, Agentic LLMs
- **Scalable Architecture & Development:** Designing and coding cloud-based e-commerce platforms
- **Software Engineering:** Data structures, algorithms, complexity analysis, object-oriented design, design patterns, middleware
- **Leadership:** Mentoring, team building, cross-functional collaboration

◆ Technical Skills

Programming Languages: Java, Python, C++, SQL, CUDA

ML/Deep Learning Frameworks: Scikit-Learn, Keras, PyTorch, **Kineto**, OpenVINO, TVM

Machine Vision ToolKit: OpenCV, CNN, Faster R-CNN, YOLO, CLIP, BLIP, GLIDE, Stable Diffusion

Graph Neural Network: NetworkX, PyTorch Geometric

MLOps & Deployment: Apache Ray, AWS SageMaker, Kubernetes, ONNX, Feature Store: FEAST

NLP ToolKit: spaCy, CoreNLP

Agentic & GenAI Frameworks: RAG, AutoGen, Lean, LangChain, LangGraph, COT, Tree of Thought

Search & Information Retrieval: Lucene, Solr, Elasticsearch, Nutch, Neural IR, DSPy

Streaming & Messaging Systems: Kafka, KSQL, Faust, Apache Storm

Data Engineering & Distributed Computing: Hadoop, Spark, PySpark, Hive, ZooKeeper, Redis

Databases: Cassandra, MongoDB, HBase, PostgreSQL, Oracle, FAISS (vector DB), WandB

Web & Middleware: Flask, Tomcat, UNIX internals

◆ Open-Source Contribution

- **Lucene-Solr Project:** Integrated location-based search and implemented a custom similarity function in the main branch.
- **Additional contributions available on [GitHub](#)** — includes projects from Udacity Nanodegrees (Self-Driving Car, Deep RL, NMT) and more.

◆ Professional Experience

EleutherAI – AI Research Scientist

May 2025 – Present

- Leading research on LLM alignment for legal reasoning, integrating the Lean theorem prover with agentic retrieval-augmented systems to formalize and verify compliance rules; exploring techniques such as RLHF and Constitutional AI to enhance transparency, traceability, and adherence to normative constraints.
- Implemented scalable evaluation systems for automated model assessment, fully integrated into **CI/CD** workflows.

GenAI Architect – KP Digital:**Jul 2022-Apr 2025**

- Designed and implemented a conversational AI chatbot for customer service, reducing support call volume by 25% and improving response accuracy. RAG Architecture to reduce hallucination
- Designed and scaled post-training pipelines to align general-purpose LLMs with target use cases using advanced techniques like RLHF, DPO, GRPO, and RLAIIF.
- Designed and deployed a conversational AI chatbot using RAG architecture, reducing support call volume by 25% and improving response accuracy.
- Built post-training pipelines to align general-purpose LLMs with domain tasks using techniques like DPO, GRPO, and RLAIIF.
- Developed NLP pipelines for literature mining to extract insights from scientific texts and biomedical databases; enabled PII anonymization, hypothesis generation, and knowledge curation.

Technologies : Azure, Kubernetes, PyTorch, T5, MedAlpaca, LLaMA2, LangChain, RLHF**Research Scientist-Meta:****Dec 2021 – Jul 2022**

- Benchmarked and optimized **data and model parallelism** for a large-scale **SparseNN Ad Ranking ML model** on cutting-edge AI accelerators and chips, improving performance and efficiency for Meta's high-traffic advertising infrastructure.
- Developed and enhanced a **personalization model** that tailored ad content to individual users preferences, leveraging real-time data and optimizing for both relevance and engagement, which led to improved user experience and ad effectiveness.
- Applied hyperparameter tuning, quantization, and QLoRA to optimize memory usage and accelerate inference while maintaining model accuracy.
- Leveraged LLVM-IR to perform advanced code analysis and optimizations, enabling efficient translation of high-level languages into optimized machine code for multiple architectures.
- Collaborated with cross-functional teams to integrate new hardware, utilizing **Python** and **CUDA** for AI model acceleration and GPU optimization.

Technologies : PyTorch, Quantization Techniques, LLM, Python, CUDA Programming**Principal NLP Engineer – CVSHealth (contract)****Jun 2021 – Nov 2021**

- Generated and curated annotated medical imaging datasets (ChestX-ray14, ISIC, TCIA, BRATS) for training diagnostic models. Leveraged DICOM processing, PyTorch, and scikit-image for scalable preprocessing and model development.
- Deployed models in clinical workflows for X-ray and MRI diagnosis, achieving 87% precision and 83% recall on held-out test data. Enabled downstream tasks such as abnormality detection and radiology triage.

Principal Machine Learning Engineer – Oto Analytics**Dec 2020 – Jun 2021**

- Architected and implemented a production-grade Feature Store using Feast, enabling consistent feature access across training and serving pipelines.

Machine Learning Engineer - ServiceNow (contract)**Jun 2020 – Nov 2020**

- Developed intent extraction and question-answering models using BERT and custom NLU pipelines to enhance customer service automation.

Principal Research Scientist (ML) - Kohl's Innovation Lab**Aug 2017 –May 2020**

- Implemented real-time object detection, segmentation, gesture recognition system non-rigid tracking with pre-trained deep learning models YOLO, SSD
- Store Shopping Intent Analysis models for people counter, Age and Gender identification.
- Optimized and deployed ML models on edge devices using Apache TVM for efficient inference.

Staff Machine Learning Engineer - Walmart Labs**May 2015 - Mar 2017**

- Built ML models for attribute extraction such as detecting brand, color, size etc. with
 - Supervised Learning B. Sequence Labelling and CRF C. **CNN** approach.
 - Relation extraction, Semantic Parsing, and knowledge graph
- Implemented Neural Network model for Item Matching System based on title and description.
 - Used pre trained Word embedding – **Glove**-contextual word representation.
- Worked in distributed data pipeline to orchestrate raw item JSON through a series of **micro services** producing a sellable item.
- High-Performance Computing (HPC) workloads: Large-Scale Machine Learning Training: Built distributed machine learning frameworks to train deep neural networks on massive datasets.

Technology: Python, Scikit Learn, TensorFlow, Keras, ZooKeeper, Kafka, Storm, Cassandra, Mongo**Senior Architect - Angie's List****Mar 2012 - Mar 2014**

- Designed and programmed Query understanding and rewriting with a heuristic Ranking module.
- Implemented a custom similarity function to enhance relevance ranking for geospatial queries, boosting revenue and retention by 25%.
- Developed and implemented a data discovery pipeline to identify, classify, and analyze structured and unstructured data, enabling improved data management and actionable insights.
- Designed, implemented a cold start solution to optimize recommendations and user experience by leveraging data analysis, machine learning models, and feature engineering for new users and items
- Implemented **Multi Objective Session based Recommender System** for Deals, Coupons

Technology: Java/J2EE, Lucene-Solr, Python, Multitask Ranking, Kea, Ling Pipe, NER, Elastic Search**Staff Software Engineer - Walmart Labs Search and Platform Team****Jun 2010 - Mar 2012**

- Designed Sentiment Analysis Network: Product Reviews data set – sentiment aware tokenizer.
- Built Meta-Search Engine with Apache Carrot, Enhanced content with annotation engine-UIMA.
- Object Tracking in video, Handwritten digit recognition, Hand gesture recognition, pose detection

Technology: Scikit Learn, SentiWordNet, SGD, Sentiment Treebank, Word2Vec, OpenCV, Flask, REST**Tech Leader – Search Science - Macys.com****May 2001-May 2010**

- Implemented automatic IR Evaluation System. **MS-MARCO** document ranking dataset, Real-Time Indexing, Search Ranking Algorithms, Auto Suggest component, Sponsored Search
- Implemented web crawler-NUTCH secure and static web pages, Faceted Search and Browse
- New Ranking algorithm for Search Relevancy Tuning, Query Understanding and rewrite,
- Implemented Deep Personalization ranking mechanism based on a user's search and click history.

Technology: Java J2EE, Tomcat, Lucene-Solr, Vector Search, **BM25, Learn to Rank, Elastic Search****◆ Education****MS in Computer Science and Applications - National Institute of Technology, India****Jun 1988****◆ Professional Development****AI Specialization – Stanford University****Jul 2009****Advanced Agentic LLM (Large Language Models) – UC Berkeley****Jun 2025****Udacity Self-Driving Car Nanodegree (Dec 2017):** Completed projects in lane detection, vehicle detection, Kalman filtering, traffic sign classification, behavioral cloning, MPC, PID control, and simulation/testing.**Technology: C++, PyTorch, OpenCV, PCL (Point Cloud Library), Mask R-CNN****◆ Publications and Presentations**

- [Detecting Cyberbullying Instances](#), published paper (Apr 2022)
- Coauthor, [Practical Convolutional Neural Networks](#), Packt Publishing (Feb 2018)
- Presentation: *Leveraging Solr for Real-time Sentiment Analysis*, Sentiment Symposium, San Francisco (May 2013)